

Intelligent systems for industry, supply chain and environment

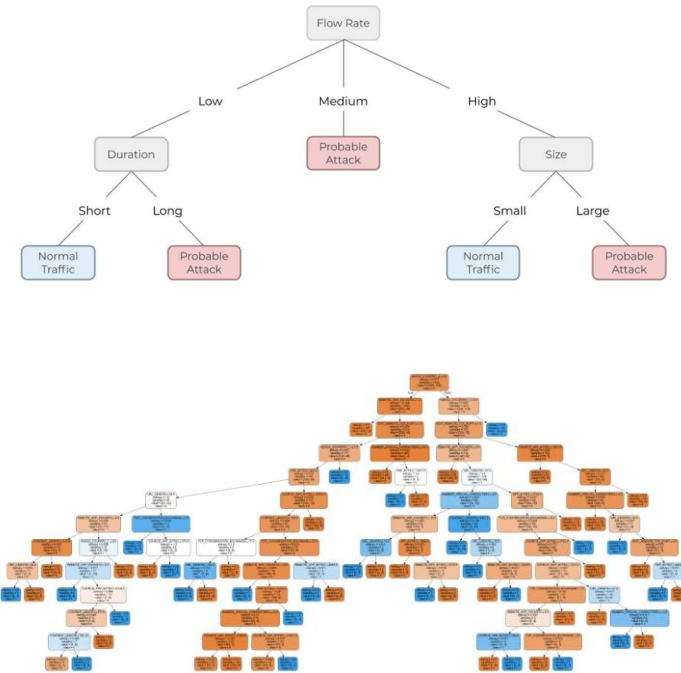
LESSON 12

Decision Tree and Random Forest
Exam simulation
of the first half of the course



Decision Trees for Security and Privacy

- Decision Trees are one of the most widely used machine learning models in security-related applications, thanks to their simplicity, interpretability, and efficiency.
- They naturally support rule-based reasoning, which is a key requirement in security, privacy, and compliance-driven systems.
 - **Explicit** and interpretable decision logic
 - **Fast inference**, suitable for real-time security systems
 - **Easy to audit**, debug, and explain to human analysts



Decision Tree: training data format X,Y

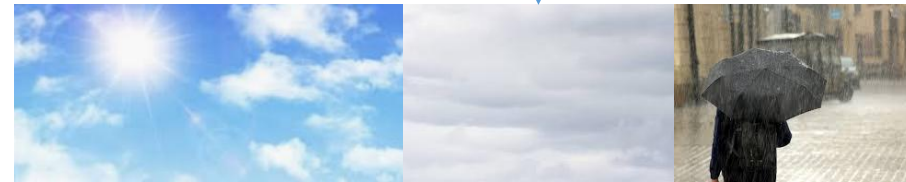
Let's go play tennis?

Dataset

Temperature	Outlook	Humidity	Windy	Played?
Mild	Sunny	80	No	Yes
Hot	Sunny	75	Yes	No
Hot	Overcast	77	No	Yes
Cool	Rain	70	No	Yes
Cool	Overcast	72	Yes	Yes
Mild	Sunny	77	No	No
Cool	Sunny	70	No	Yes
Mild	Rain	69	No	Yes
Mild	Sunny	65	Yes	Yes
Mild	Overcast	77	Yes	Yes
Hot	Overcast	74	No	Yes
Mild	Rain	77	Yes	No
Cool	Rain	73	Yes	No
Mild	Rain	78	No	Yes

X

Y

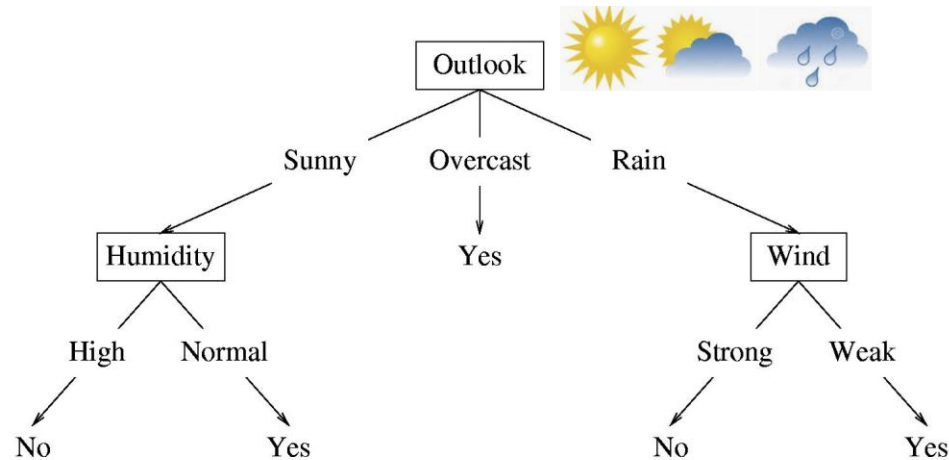


Available data

Decision Trees are perfect for tabula data

Decision Tree: Hypothesis space

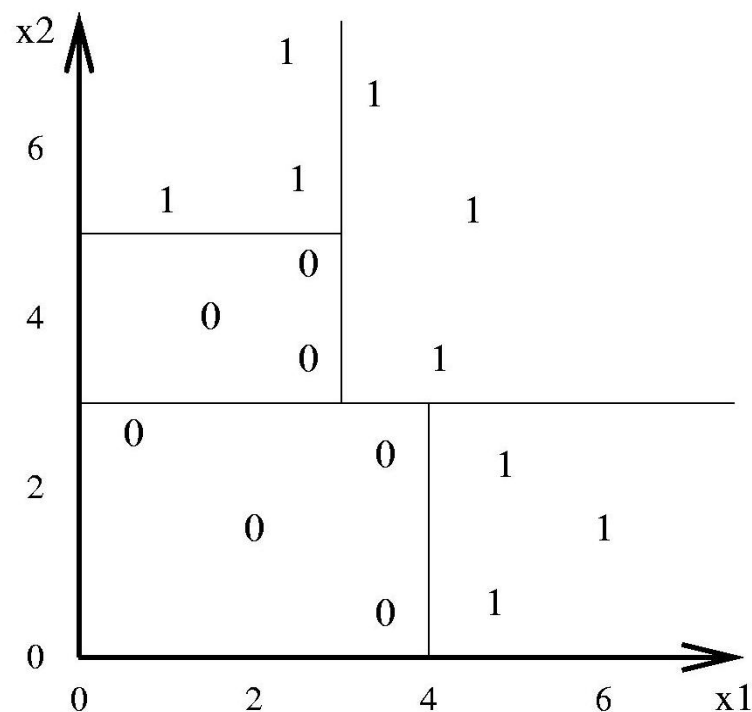
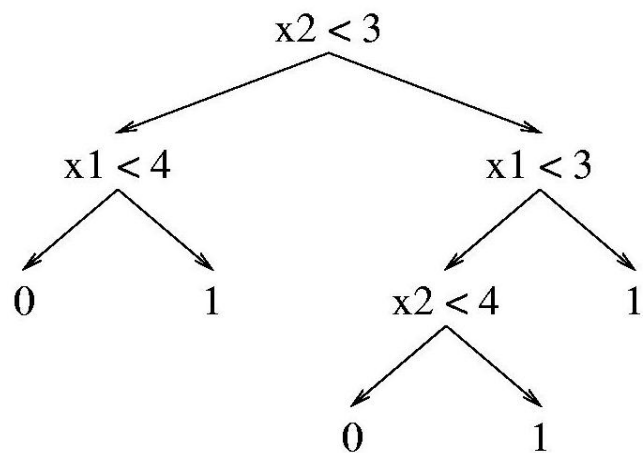
- **Internal nodes** test the value of particular features x_j and branch according to the results of the test.
- **Leaf nodes** specify the class $h(\mathbf{x})$.



Suppose the features are **Outlook** (x_1), **Temperature** (x_2), **Humidity** (x_3), and **Wind** (x_4). Then the feature vector $\mathbf{x} = (\text{Sunny}, \text{Hot}, \text{High}, \text{Strong})$ will be classified as **No**. The **Temperature** feature is irrelevant.

Decision Tree Decision Boundaries

Decision trees divide the feature space into axis-parallel rectangles, and label each rectangle with one of the K classes.



Decision Tree: Explainability

- **Decision Trees are inherently interpretable models.**

Their structure makes explicit *how* and *why* a decision is taken, by organizing decisions as a sequence of simple, human-readable rules.

- Each internal node corresponds to a **feature-based decision**
- Each path from root to leaf represents a **decision rule**
- Features that do not contribute to reducing uncertainty are **automatically discarded**

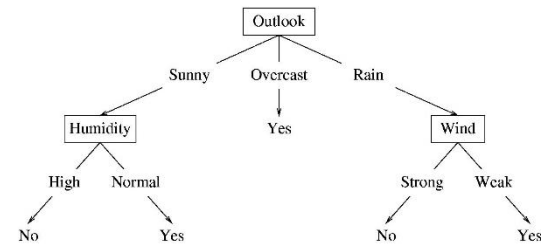
- **Example (Play Tennis dataset)**

- In our example, the feature *Temperature* does not improve decision quality and hence it is not present in the tree



Dataset

Temperature	Outlook	Humidity	Windy	Played?
-------------	---------	----------	-------	---------



Binary VS non Binary trees

Binary trees

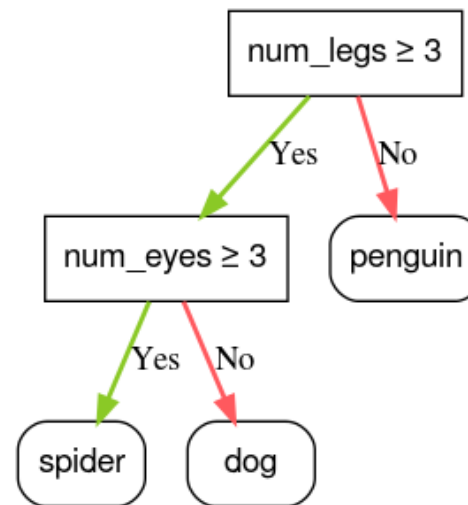
- Each internal node splits the data into **two branches**
- Simpler structure and easier pruning

Non-binary trees

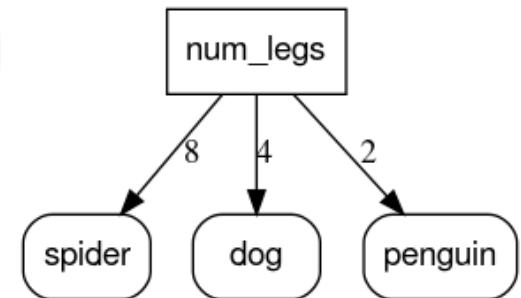
- Nodes may have **multiple branches**
- Often used with categorical features
- Shallower trees but more complex decision nodes

Trade-off

- Binary trees → higher depth, simpler decisions
- Non-binary trees → lower depth, higher node complexity

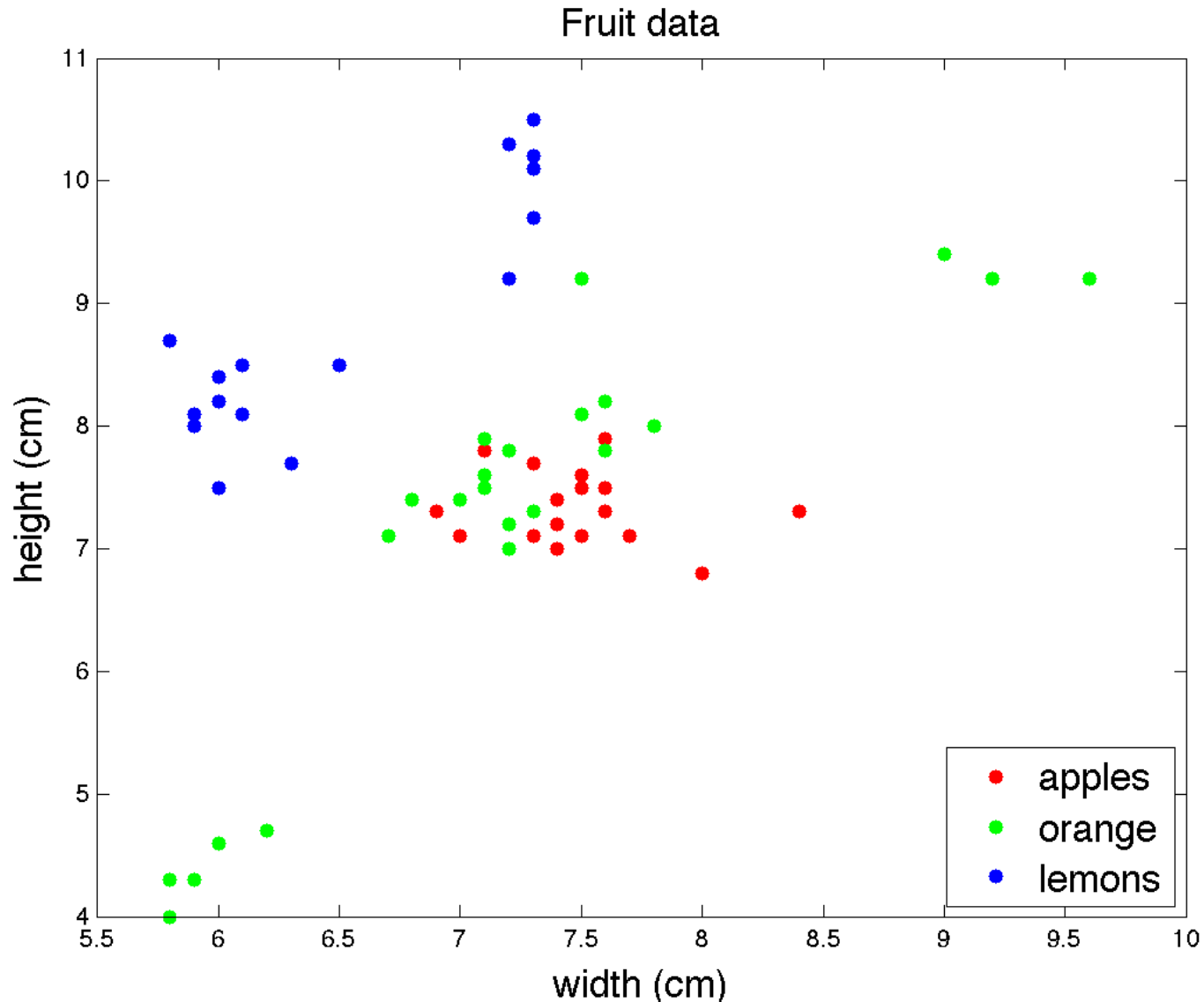


A binary decision tree

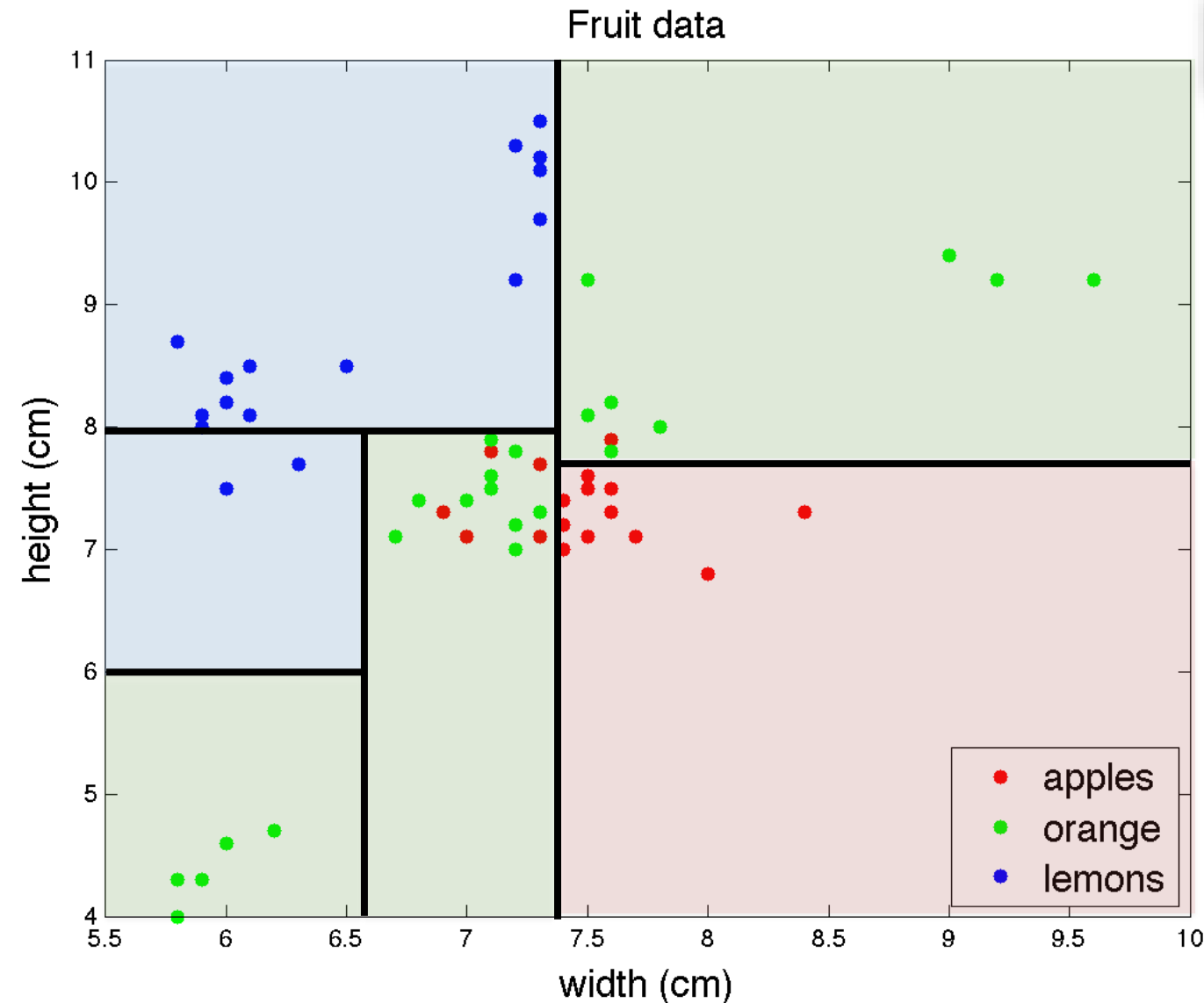


A non-binary decision tree

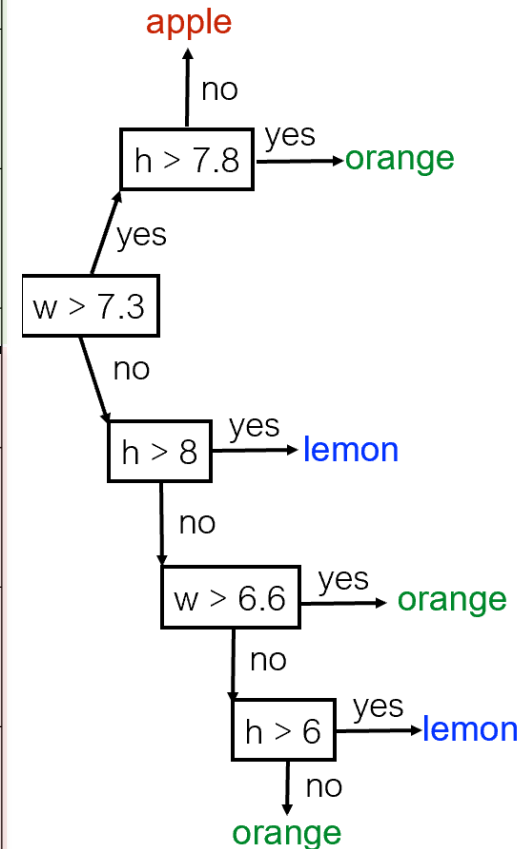
Review the «fruit data» case for a Decision Tree



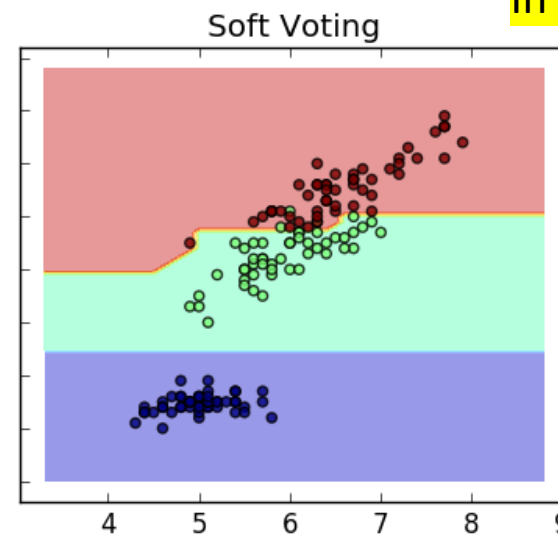
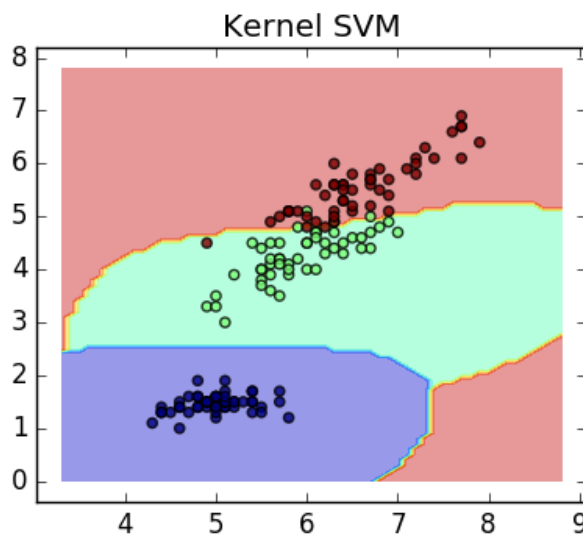
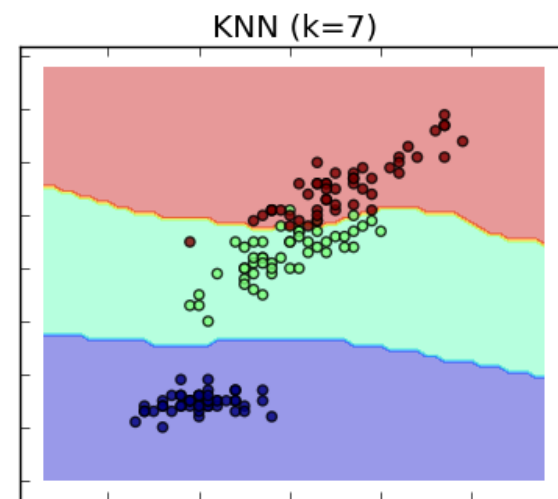
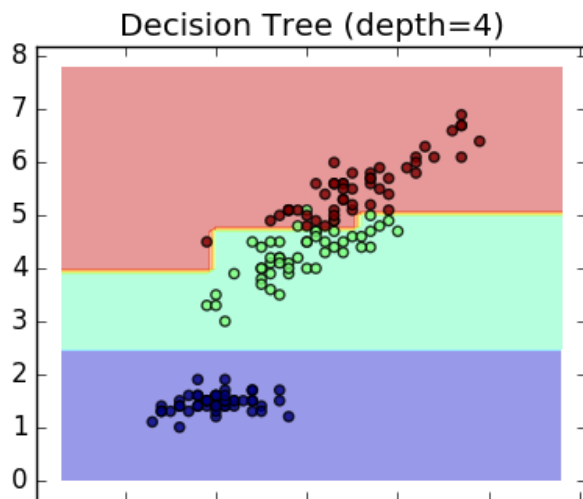
Decision boundaries of a Decision Tree



The decision boundaries are axis aligned for DT

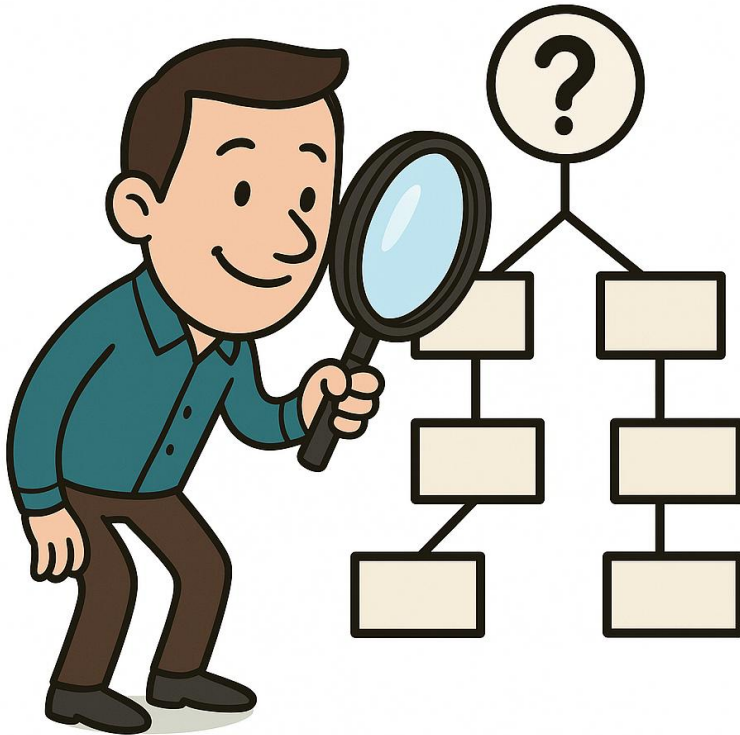


Comparing the decision boundaries of different models



In the next lessons

How to check the knowledge in the tree?



- 1) Depth of the tree
- 2) Information in nodes
 - Gini Impurity
 - Entropy

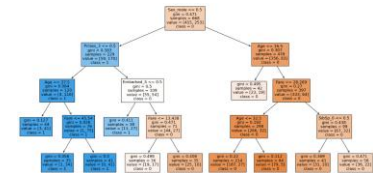
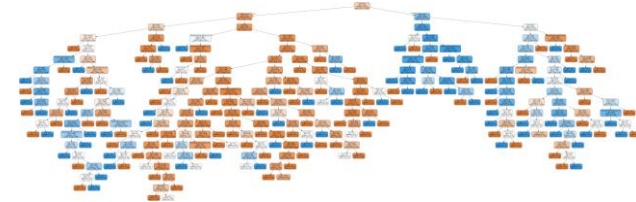
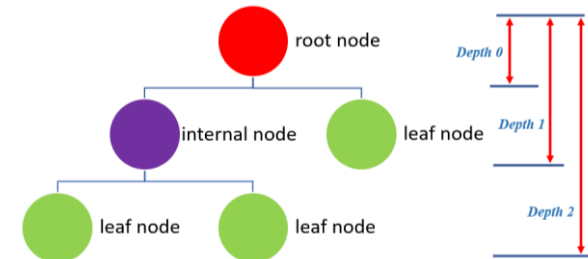
How to check the knowledge in the tree?

1) Depth of the tree



The depth tells us how many questions (decisions) the tree must ask, from the root to a final answer (leaf).

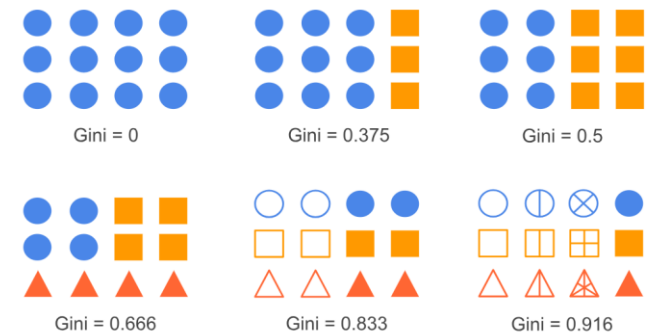
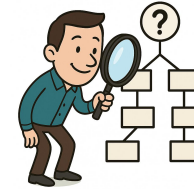
- A **deeper tree** means more complex logic. It may capture more detail but risks overfitting.
- A **shallow tree** is simpler faster and easier to interpret, but might miss important patterns.
- The **depth** is like the number of steps it takes for our little detective to reach a decision!



Decision Tree are very very fast.
(only few ifs!)

How to check the knowledge
in the tree?

Gini Impurity



- **Node Purity** in Decision Trees

Gini Impurity measures the chance that a randomly chosen element would be incorrectly classified if it were labeled based on the distribution of classes in a node.

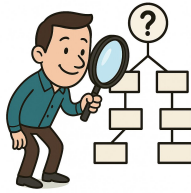
- It is used to **evaluate splits** in Decision Tree algorithms like **CART**.

$$Gini(t) = 1 - \sum_{i=1}^C p_i^2$$

- t : node
- C : number of classes
- p_i : proportion of class i in the node

Scenario	Gini Value	Meaning
All samples same class	0.0	Pure node
50% Class A, 50% Class B	0.5	Maximum impurity
70% A, 30% B	0.42	Partial impurity

How to check the knowledge in the tree? Entropy



- Entropy measures the uncertainty in a node.
 - It quantifies how mixed the classes are
 - higher entropy means more confusion.
- It is used to evaluate splits in Decision Tree algorithms like ID3 and C4.5

$$Entropy(t) = - \sum_{i=1}^C p_i \log_2 p_i$$

Entropy comes from **information theory** (Shannon, 1948), and the **logarithm base** defines the **unit of information**:
In this context, *information* is actually a bad thing — it means the node has too much uncertainty

- t : node
- C : number of classes
- p_i : proportion of class i in the node

Entropy 2)

$$H = - \sum_{i=1}^K p_i \log_2(p_i)$$

- Entropy measures uncertainty in class labels.
- Minimum Entropy
 - Entropy is minimized when there is no uncertainty.
This happens when all samples belong to a single class.
- Maximum Entropy (Decision Trees)
 - Entropy measures uncertainty in class labels. It reaches its maximum when all classes are equally likely.
 - Hmax = 1 for the binary case.
 - Example (3 Classes: Apple, Pear, Banana)
- Why log2? Because the entropy is measured in bits! (for p(1)=50% → H = 1 bit)

$$p_j = 1 \quad \text{for one class } j, \quad p_i = 0 \quad \forall i \neq j$$

(By convention: $\lim_{p \rightarrow 0^+} p \log p = 0$)

$$H_{\min} = 0$$

$$p = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

$$H = -3 \cdot \frac{1}{3} \log_2 \left(\frac{1}{3} \right) = \log_2(3) \approx 1.585$$

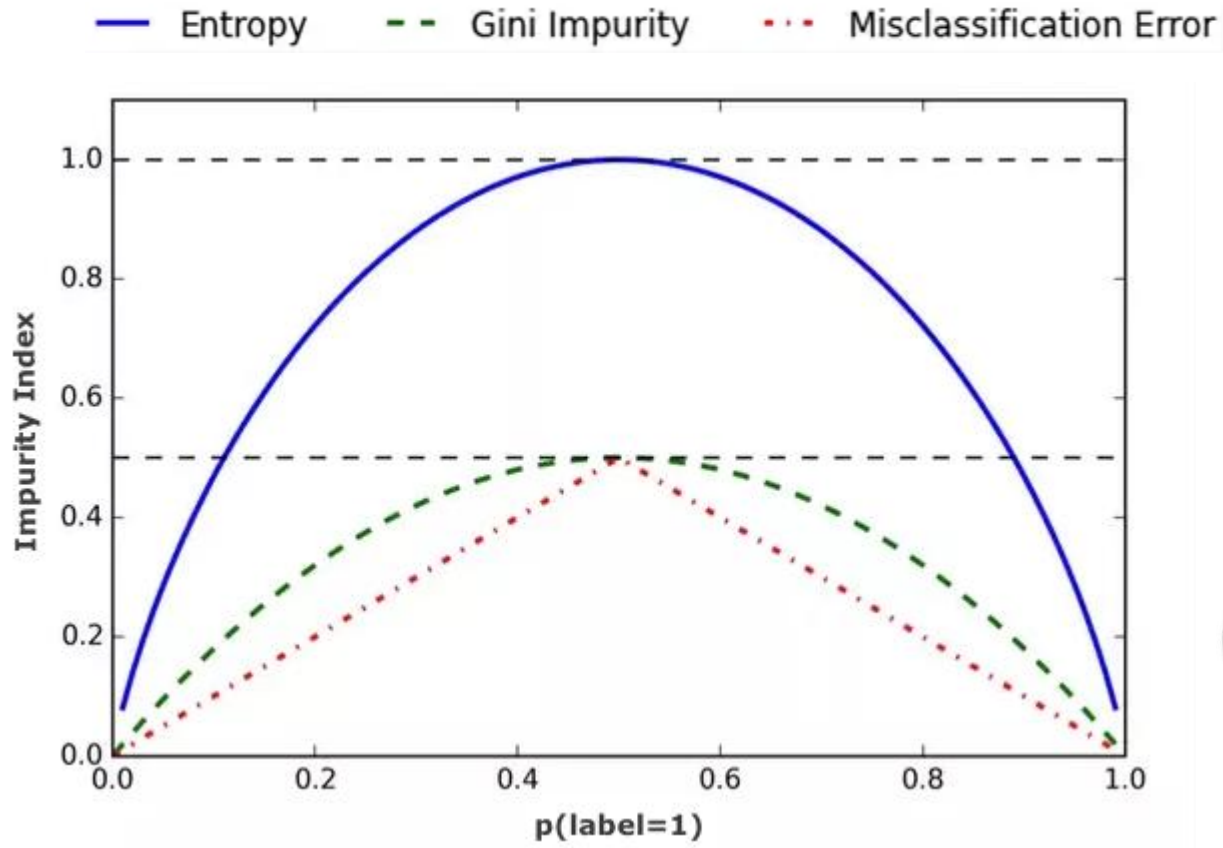
Entropy in Machine Learning

$$H = - \sum_{i=1}^K p_i \log_2(p_i)$$

A very important concept applied to

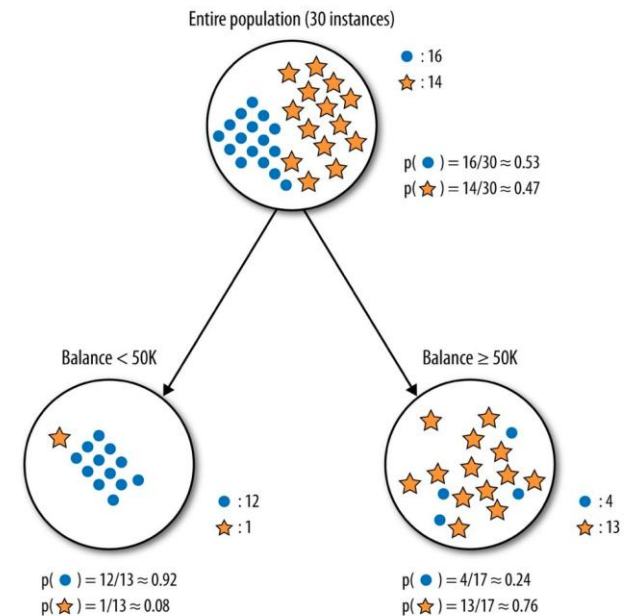
- **Decision Trees**
 - Used to decide the best feature to split the data (ID3, C4.5 algorithms)
- **Information Gain**
 - Entropy is used to compute how much uncertainty is reduced by a split
- **Feature Selection**
 - Helps rank features by how much information they provide
- **Unsupervised Learning** (e.g. clustering)
 - Measures how well clusters separate the data (e.g., in mutual information)
- **Loss Functions**
 - Cross Entropy Loss
- **Regularization**
 - Label Smoothing Used to prevent overconfidence
- Etc..

Entropy and Gini

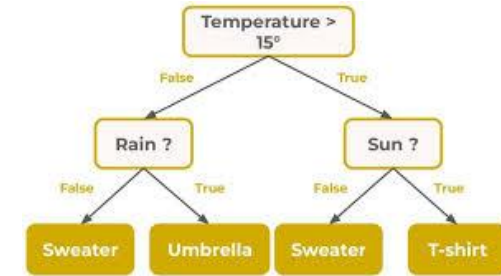


- p_i : proportion of class i in the node

For a BINARY Class problem!



Decision Tree in brief



Common Training Algorithms: ID3, C4.5, CART.

Advantages

- **Interpretability:** Simple to understand and visualize.
- **Versatility:** Can handle both numerical and categorical data.
- **Minimal Data Preparation:** Requires little to no data normalization.
- **Non-Parametric:** No assumptions about data distribution (e.g., Gaussian)

High Interpretability

Handles Mixed Data

Efficient for Small Data

Disadvantages

- **Overfitting:** Complex trees can model noise, reducing generalization.
- **Instability:** Small data changes can result in a completely different tree.
- **Bias with Imbalanced Data:** May be biased towards dominant classes

Prone to Overfitting

Sensitive to Noise

Instability

Fine tuning

- **Pruning:** Reduces tree size to prevent overfitting.
- **Feature Selection:** Important for improving model performance.
- **Ensemble Methods:** Combining multiple trees (e.g., Random Forests) enhances accuracy.

From one Tree to one Forest

Decision Tree Instability

Decision trees are highly sensitive to the training data. Small changes in the dataset can produce very different trees.

Why?

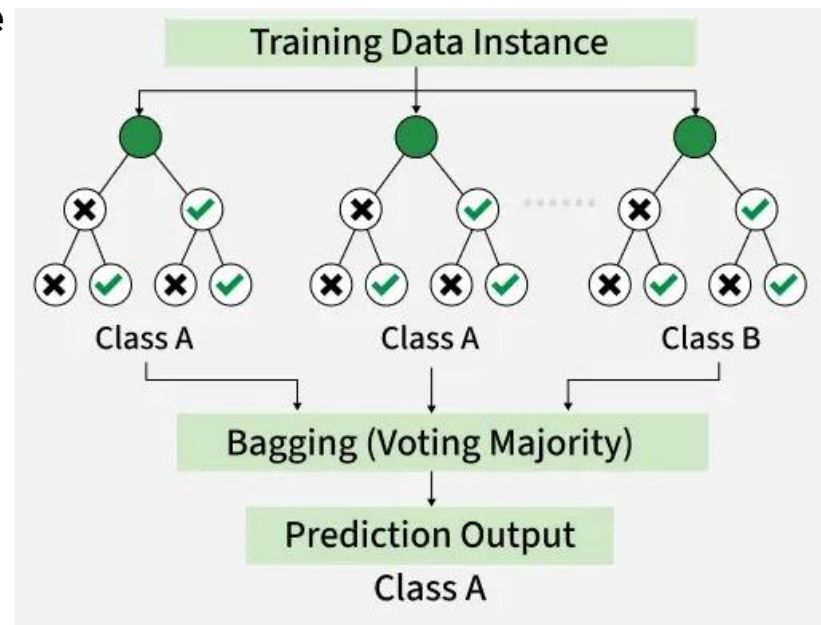
Because A decision tree is built using a **greedy, local optimization**

By sampling different subsets of the dataset:

- multiple trees can be trained
- each tree captures a different view of the data

Decision Forest

- The same training instance is evaluated by multiple decision trees. Each tree produces an independent prediction.
- Trees are trained on **different bootstrap samples** of the training data
- Due to data and feature randomness, **trees make different decisions**
- Each tree outputs a predicted class (e.g., *Class A* or *Class B*)
- **Bagging (Bootstrap Aggregating)**
 - combines predictions from all trees
 - **classification**: majority voting
 - **regression**: average prediction
- **Final prediction**
 - the class receiving the majority of votes is selected
 - individual tree errors tend to cancel out



Decision Tree vs Random Forest: When to Use Which

- **Decision Trees and Random Forests offer different trade-offs between interpretability and performance.**
The choice depends on the task requirements and constraints.

- **Use a Decision Tree when**

- model interpretability is critical
- decisions must be explained or audited
- the dataset is small or simple
- rules need to be inspected or validated by humans

Choose a tree
to explain decisions

- **Use a Random Forest when**

- predictive accuracy and robustness are the priority
- the dataset is noisy or complex
- overfitting must be reduced
- some loss of interpretability is acceptable

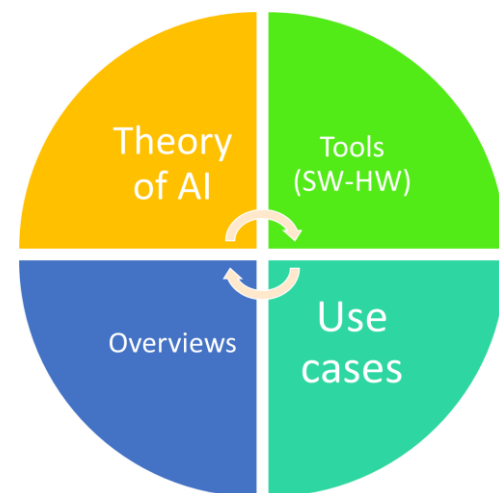
Choose a forest to
improve performance.



Tools (SW-HW)

Decision Tree in Py

Let's start with our first ML model



Wine inspector!

Automating Wine Quality Control with Decision Trees
(This is a didactic example!)

In prestigious wine cellars the composition of a wine is a matter of **heritage, science, and precision**. For example, of a **Sassicaia bottle can be priced over €400**

Blending the right **cultivars** (grape varieties) determines not only the **taste**, but also the **identity and certification** of the wine.

For example the **Chianti**: by regulation, it must include at least **70% Sangiovese**, and may be blended with Canaiolo, Colorino or even Cabernet Sauvignon

Why a Intelligent System capable to identify the cultivar can help?

- Manually checking grape composition is **slow and subjective**
- Mislabeling or adulteration affects **brand, price, and legality**
- High-end wineries want **automated systems to detect and classify** wines/cultivar from chemical or sensor data



Tenuta San
Guido - Bolgh...

€440.00

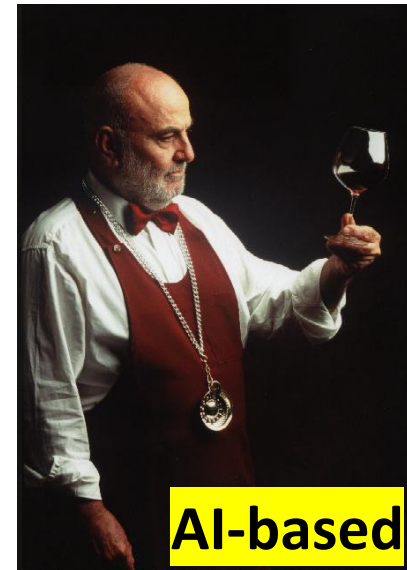
Tannico

Free shipping

Automating Wine Quality Control with Decision Trees

Using Decision Trees, we can:

- Analyze wine features (like alcohol, flavonoids, proline...)
- Predict the likely grape blend or cultivar class
- Support automatic labeling, fraud detection, or production monitoring



PROCEDURE

- 1) **Load** a proper dataset
- 2) **Train** the tree
- 3) **Use** the tree to make new decisions



Load the dataset



Dataset Characteristics

- Number of instances: 178
- Number of attributes: 13 numeric, predictive attributes + 1 class label
- Missing values: None
- Class distribution:
 - class_0: 59 samples
 - class_1: 71 samples
 - class_2: 48 samples

**Wine**
Donated on 6/30/1991

Using chemical analysis to determine the origin of wines

Dataset Characteristics	Subject Area	Associated Tasks
Tabular	Physics and Chemistry	Classification
Feature Type	# Instances	# Features
Integer, Real	178	13

Dataset Information

For what purpose was the dataset created?

test

Additional Information

These data are the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. The analysis determined the quantities of 13 constituents found in each of the three types of wines.

I think that the initial data set had around 30 variables, but for some reason I only have the 13 dimensional version. I had a list of what the 30 or so variables were, but a.) I lost it, and b.), I would not know which 13 variables are included in the set.

The attributes are (dontated by Riccardo Leardi, riclea@anchem.unige.it)

- 1) Alcohol
- 2) Malic acid
- 3) Ash
- 4) Alcalinity of ash
- 5) Magnesium
- 6) Total phenols
- 7) Flavanoids
- 8) Nonflavanoid phenols
- 9) Proanthocyanins
- 10)Color intensity
- 11)Hue
- 12)OD280/OD315 of diluted wines
- 13)Proline

In a classification context, this is a well posed problem with "well behaved" class structures. A good data set for first testing of a new classifier, but not very challenging.

SHOW LESS ^

<https://archive.ics.uci.edu/dataset/109/wine>

Classes



The Classes in the UCI Wine Dataset

The dataset includes wines produced by **three different cultivars** (grape varieties) grown in the same region of Italy. The wines were chemically analyzed and labeled as:

Class Label	Description	Cultivar (presumed)
0	Cultivar 1 (Class_0)	Barolo
1	Cultivar 2 (Class_1)	Grignolino
2	Cultivar 3 (Class_2)	Barbera

Load the dataset

```
# Import required libraries
import pandas as pd
from sklearn.datasets import load_wine
from sklearn.tree import DecisionTreeClassifier #the real model
import matplotlib.pyplot as plt
from sklearn import tree # the functions like tree.plot_tree()
```

The variable `wine` is a Bunch object from scikit-learn.

A Bunch is a special data structure that works like a dictionary but also allows attribute-style access.

The `wine` object contains:

- `.data` : a NumPy array of shape `[n_samples, n_features]` with the raw feature values
- `.target` : a 1D array of class labels (0, 1, 2)
- `.feature_names` : a list of the 13 feature names
- `.target_names` : a list of class names (`['class_0', 'class_1', 'class_2']`)
- `.DESCR` : a full textual description of the dataset
- `.frame` : (optional) a pandas DataFrame version of the dataset (if `as_frame=True` is used)

Checking the vars

```
In [10]: y
```

[illegible]

```
In [11]: X
```

```
array([[1.423e+01, 1.710e+00, 2.430e+00, ..., 1.040e+00, 3.920e+00,
        1.065e+03],
       [1.320e+01, 1.780e+00, 2.140e+00, ..., 1.050e+00, 3.400e+00,
        1.050e+03],
       [1.316e+01, 2.360e+00, 2.670e+00, ..., 1.030e+00, 3.170e+00,
        1.185e+03],
```

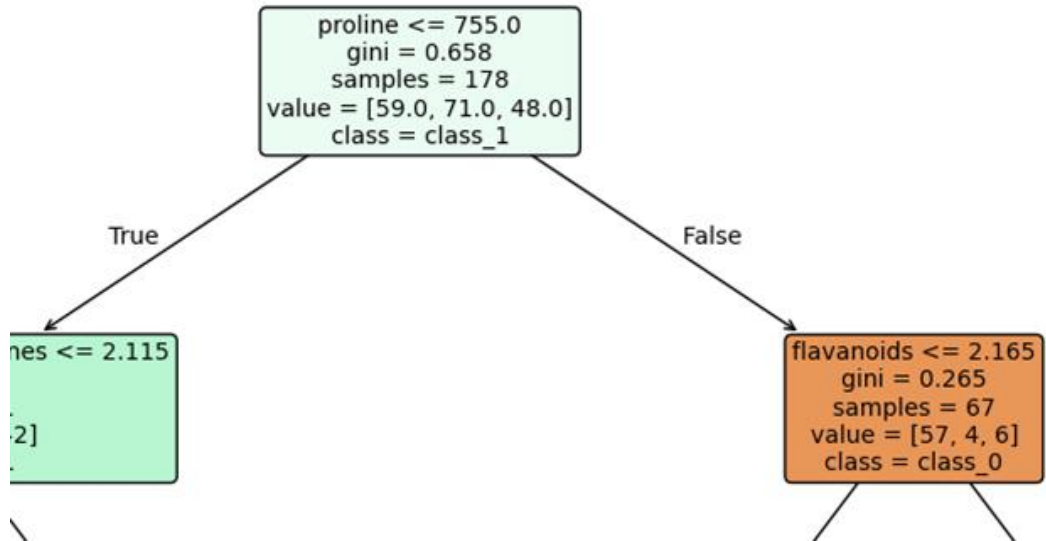
Initialization and Training

```
# 2. Train the Decision Tree Classifier
clf = DecisionTreeClassifier(max_depth=3) # Limit depth for simplicity
clf.fit(X, y)
```

```
# Visualize the tree
plt.figure(figsize=(15,10))
tree.plot_tree(clf,
                feature_names=wine.feature_names,
                class_names=wine.target_names,
                filled=True,
                rounded=True,
                fontsize=10)

plt.show()
```

The trained tree (3 levels)



$$Gini(t) = 1 - \sum_{i=1}^C p_i^2$$

feature <= threshold

The splitting condition at that node

gini

Gini impurity of the node — 0 means perfectly pure

samples

How many training samples reached this node

value

A list: the **number of samples per class** (e.g., [0, 1, 39])

class

The **predicted class** for that node (the majority class in value)

Class distribution:

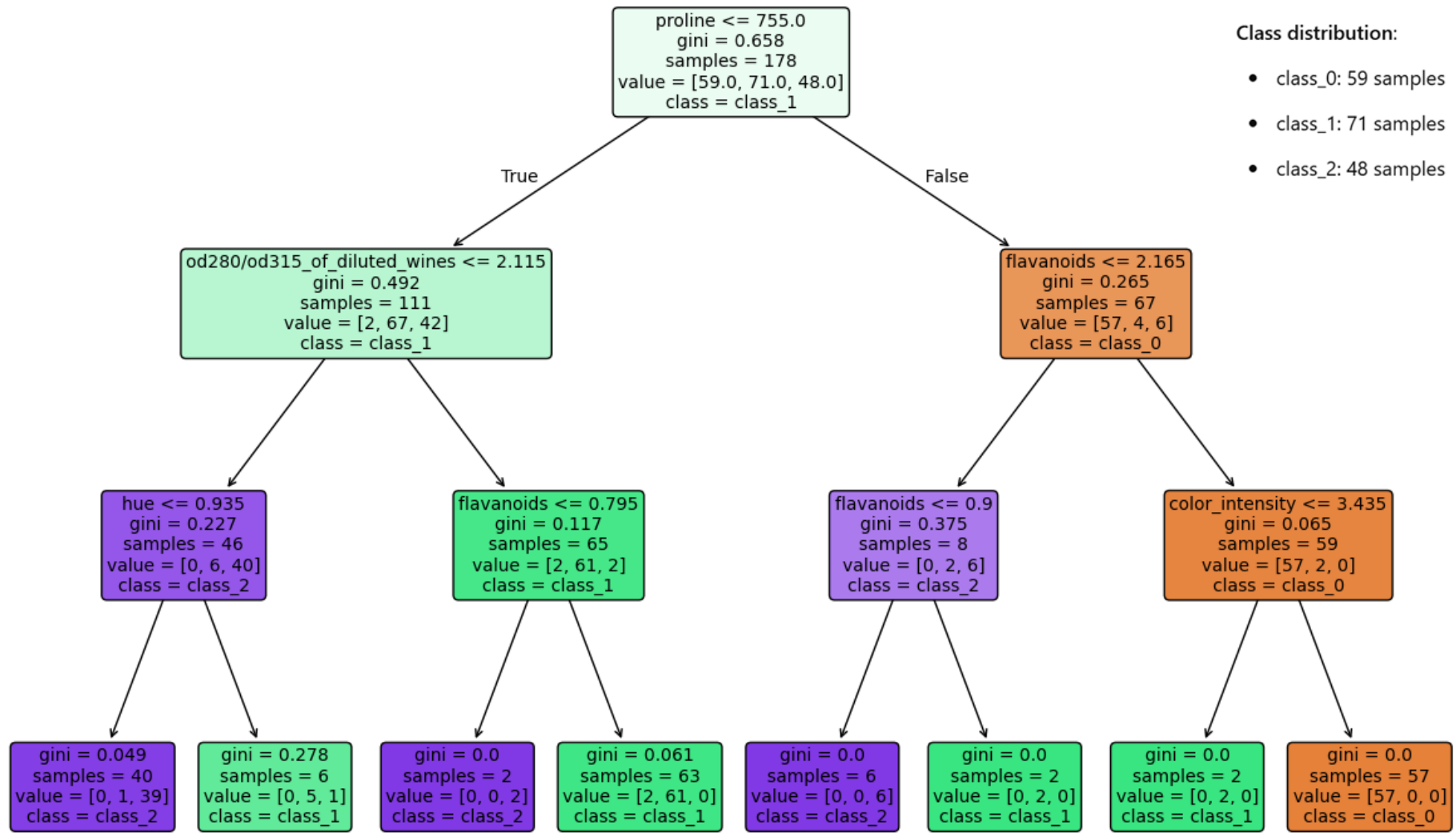
- class_0: 59 samples
- class_1: 71 samples
- class_2: 48 samples

The trained tree (3 levels)

$$Gini(t) = 1 - \sum_{i=1}^C p_i^2$$

Class distribution:

- class_0: 59 samples
- class_1: 71 samples
- class_2: 48 samples

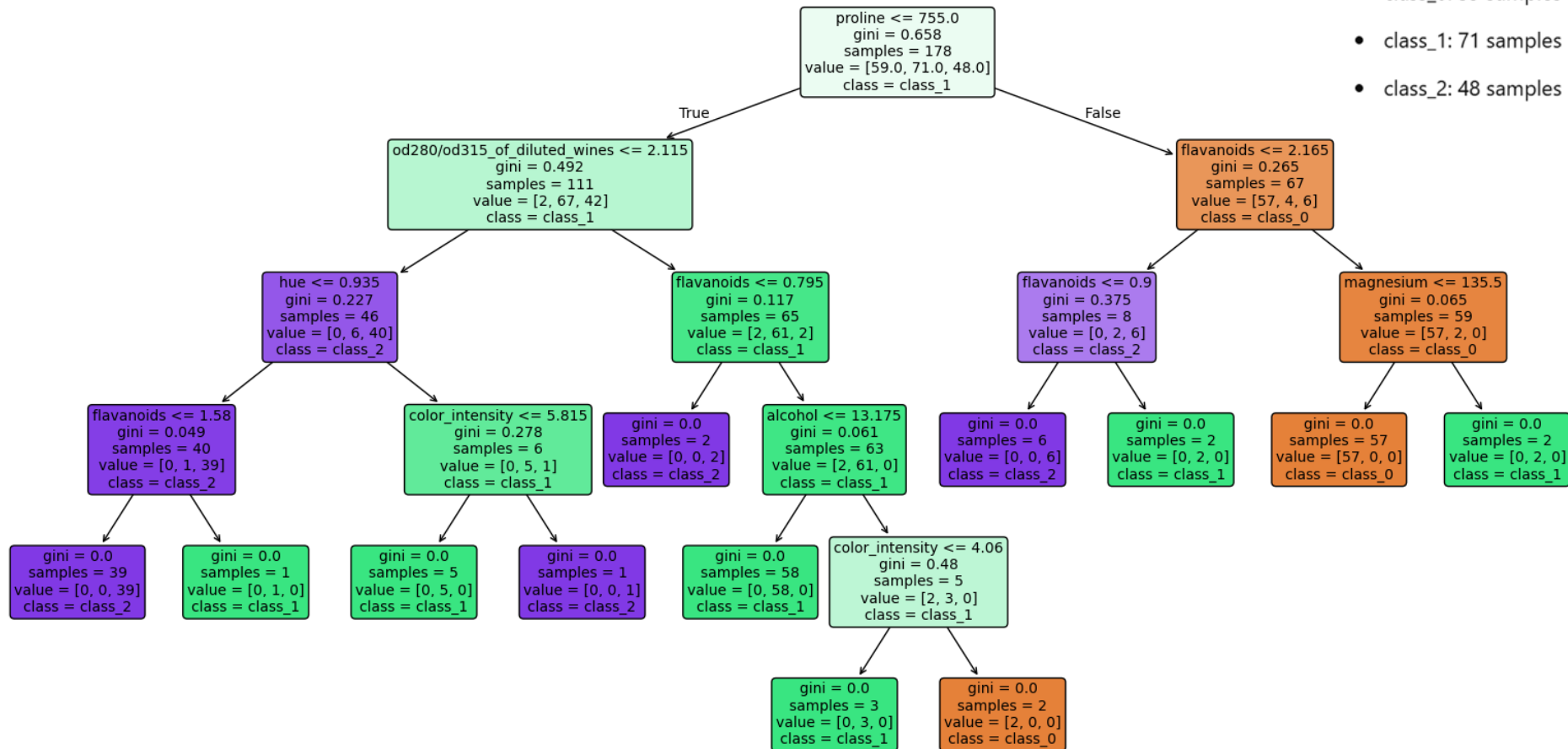


The trained tree (5 levels)

$$Gini(t) = 1 - \sum_{i=1}^C p_i^2$$

Class distribution:

- class_0: 59 samples
- class_1: 71 samples
- class_2: 48 samples



With 5 levels the tree is now able to learn perfectly the training dataset (Gini = 0 in every leaf)

Use the tree (training sample)

```
# 3. Example: Classify an existing data point from the dataset
# Let's take the first sample (index 0) as an example
sample_existing = X[0].reshape(1, -1) # Reshape for prediction
prediction_existing = clf.predict(sample_existing)
print("\nExample classification of an existing wine (first sample):")
print("Features:", dict(zip(wine.feature_names, X[0])))
print("Predicted class:", wine.target_names[prediction_existing[0]])
```

Use the tree (training sample)

```
# 3. Example: Classify an existing data point from the dataset
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print("Features:", dict(zip(wine.feature_names, X[0])))
print("Predicted class:", wine.target_names[prediction_existing[0]])
```

```
Example classification of an existing wine (first sample):
Features: {'alcohol': 14.23, 'malic_acid': 1.71, 'ash': 2.43,
'alkalinity_of_ash': 15.6, 'magnesium': 127.0, 'total_phenols':
2.8, 'flavanoids': 3.06, 'nonflavanoid_phenols': 0.28,
'proanthocyanins': 2.29, 'color_intensity': 5.64, 'hue': 1.04,
'od280/od315_of_diluted_wines': 3.92, 'proline': 1065.0}
Predicted class: class_0
```

Use the tree with a new sample



Barolo Riserva 'Rocche
dell'Annunziata'...

€350.00

Callmewine IT

Free shipping

Special offer

```
# 4. Example: Classify a new, unseen wine sample
# Create a plausible new wine sample (values are reasonable based on dataset ranges)
new_wine = [[13.5, 2.0, 2.4, 20.0, 100.0, 2.8, 3.0, 0.3, 1.6, 5.0, 1.0, 3.0, 1000.0]]
prediction_new = clf.predict(new_wine)
print("\nClassification of a new, unseen wine:")
print("Features:", dict(zip(wine.feature_names, new_wine[0])))
print("Predicted class:", wine.target_names[prediction_new[0]])
```

Use the tree with a new sample



Barolo Riserva 'Rocche
dell'Annunziata'...

€350.00

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```
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prediction_new = clf.predict(new_wine)
print("\nClassification of a new, unseen wine:")
print("Features:", dict(zip(wine.feature_names, new_wine[0])))
print("Predicted class:", wine.target_names[prediction_new[0]])
```

```
Classification of a new, unseen wine:
Features: {'alcohol': 13.5, 'malic_acid': 2.0, 'ash': 2.4,
'alkalinity_of_ash': 20.0, 'magnesium': 100.0, 'total_phenols':
2.8, 'flavanoids': 3.0, 'nonflavanoid_phenols': 0.3,
'proanthocyanins': 1.6, 'color_intensity': 5.0, 'hue': 1.0, 'od280/
od315_of_diluted_wines': 3.0, 'proline': 1000.0}
Predicted class: class_0
```

Decision tree in python

```
rules = export_text(dtree, feature_names=feature_names)
print(rules)
```

Decision Tree rules:

```
|--- petal width (cm) <= 0.80
|   |--- class: 0
|--- petal width (cm) > 0.80
|   |--- petal width (cm) <= 1.65
|       |--- petal length (cm) <= 5.00
|           |--- class: 1
|           |--- petal length (cm) > 5.00
|               |--- class: 2
|       |--- petal width (cm) > 1.65
|           |--- petal length (cm) <= 4.85
|               |--- class: 2
|           |--- petal length (cm) > 4.85
|               |--- class: 2
```

Random Forest in python

```
from sklearn.ensemble import RandomForestClassifier
```

```
# --- Create the Random Forest ---  
rf = RandomForestClassifier(  
    n_estimators=100,      # number of trees in the forest  
    criterion="entropy",   # same split criterion as the single tree  
    max_depth=None,       # let trees grow fully  
    bootstrap=True,       # sampling with replacement  
    random_state=1,       # use all cores (Colab-friendly)  
    n_jobs=-1  
)  
  
# Train the forest  
rf.fit(X_train, y_train)
```

Random Forest in python

```
# =====  
# 5. Calling the forest  
# =====  
  
# --- Predict a single sample ---  
sample = X_test[0].reshape(1, -1)  
pred_class = rf.predict(sample)[0]  
  
print("\nSingle sample prediction by Random Forest:")  
print("Predicted class:", class_names[pred_class])  
  
# --- Access individual trees ---  
print("\nNumber of trees in the forest:", len(rf.estimators_))  
  
first_tree = rf.estimators_[0]  
second_tree = rf.estimators_[1]
```

Trees and forest in ML

Decision Trees and Random Forests are not only powerful classification models, but also valuable tools for understanding data.

By learning a hierarchy of decision rules, trees show:

- **how data can be separated** into homogeneous groups,
- **which features are used to discriminate classes**, and
- **where decision boundaries naturally emerge.**

Random Forests extend this idea by combining many different trees:

- each tree captures a **different view of the data**,
- the ensemble provides **more stable and robust decisions**,
- feature importance highlights **which variables matter most on average.**

Main points



- Decision Tree
 - a classical and explainable model
 - Decision boundaries
 - Gini Impurity
 - Entropy
 - Training, plotting and using a Decision Tree
- Random Forest
 - Less explainability but more robust decisions



EXAM SIMULATION

½ of the course

To TEST your preparation

(do not use the simulation to study)



Previous simulation



Simulation @Lesson #4

about questions regarding the first 4 lessons

Simulation @Lesson #7

about questions regarding the first 7 lessons

In this simulation we will focus on Lessons #4-12

Please remember



Multiple-choice test:

- 1 single correct answer $\rightarrow weight = 1$
- N wrong answers $\rightarrow weight = 0$
- No negative weights

1. No penalties for not answering
 $\rightarrow$ OK BUT IT'S BETTER TO TRY TO ANSWER
2. **DO NOT LOOK FOR THE RIGHT ANSWER, EXCLUDE THE WRONG ONES**
3. DOUBTS? $\rightarrow$ Choose the answer that "looks less wrong" to you

In brief: how to answer?

HINT:

DON'T LOOK FOR THE RIGHT ANSWER!!!

STEP 0:

- Read the text paying attention to any negations or indications (not increasing, less than,...)

STEP 1

- ~~Rule out/discard~~ answers that seem wrong using reasoning, a counterexample, a notion, a sizing/dimensioning calculation

STEP 2

- Select the answer that **SEEMS LESS WRONG THAN THE OTHERS**

Simulation Preparation:

Print this slide (1 correct answer=1 point)

#	Correct? 1/0	Notes	#	Correct? 1/0	Notes
1			11		
2			12		
3			13		
4			14		
5			15		
6			16		
7			17		
8			18		
9			19		
10			20		

1



On your mark,
get set, ...



Intelligent Systems: data sources



Considering IoT devices as source of data for **external** intelligent systems (IS is not intended to be embedded into the IoT device), what kind of IoT devices can be really used?

1. Passive data IoT devices
2. Active data IoT devices
3. Dynamic data IoT devices
4. All of the above
5. None of the above

A: Intelligent Systems: data sources



Considering IoT devices as source of data for **external** intelligent systems (IS is not intended to be embedded into the IoT device), what kind of IoT devices can be really used?

1. Passive data IoT devices
2. Active data IoT devices
3. Dynamic data IoT devices
4. All of the above
5. None of the above

Reminder: passive data does not mean a passive data/IoT application. Indeed, since the sensors need to be managed, the application must take on the logic needed to do so.

2



On your mark,
get set, ...



Design of Intelligent Systems



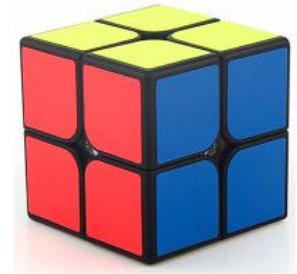
Referring to the class discussion, the (correct) design practice for neural networks considers

1. Start with deep learning models since they are the cutting edge and most advanced technology we have now
2. Start with deep learning models since they are the cutting edge and most advanced technology we have now, and then use classical method as reference
3. Start with simple neural networks before to consider deep learning models

A: Design of Intelligent Systems

Referring to the class discussion, the (correct) design practice for neural networks considers

1. Start with deep learning models since they are the cutting edge and most advanced technology we have now
2. Start with deep learning models since they are the cutting edge and most advanced technology we have now, and then use classical method as reference
3. Start with simple neural networks before to consider deep learning models



Applications
of the Occam's razor

3



On your mark,
get set, ...



Data Preprocessing: missing data



The missing values can also be occupied by computing mean, mode or median of the observed given values.

1. This is very unusual and not common in practice
2. This is a very simple and effective solution in case the learning method is not capable to deal with missing data
3. This is not possible, since that is just descriptive statistics about the features, and cannot be used to fill missing data

A: Data Preprocessing: missing data



The missing values can also be occupied by computing mean, mode or median of the observed given values.

1. This is very unusual and not common in practice
2. This is a very simple and effective solution in case the learning method is not capable to deal with missing data
3. This is not possible, since that is just descriptive statistics about the features, and cannot be used to fill missing data

4



On your mark,
get set, ...



Generalization and dataset preparation



Referring to the class discussion on data leakage what is the worst situation?

1. The unwanted leakage of data from test dataset to training data set
2. The unwanted leakage of data from training dataset to test data set
3. None of the above since transferring data from test and/or training dataset is normal when the accuracy of the model is tested



A: Generalization and dataset preparation



Referring to the class discussion on data leakage what is the worst situation?

1. The unwanted leakage of data from test dataset to training data set
 - Since large model can learn perfectly/overfit training data, if you are subtracting data to the generalization test, making the situation more **optimistic**
2. The unwanted leakage of data from training dataset to test data set
3. None of the above since transferring data from test and/or training dataset is normal when the accuracy of the model is tested



5



On your mark,
get set, ...



Generalization and dataset preparation



An additional information can allow the model to learn or know something that it otherwise would not know and in turn invalidate the estimated performance of the model being constructed. This is called

1. Data leakage
2. Data pre-processing
3. Data harmonization
4. Data wrangling

A: Generalization and dataset preparation



An additional information can allow the model to learn or know something that it otherwise would not know and in turn invalidate the estimated performance of the model being constructed. This is called

1. Data leakage
2. Data pre-processing
3. Data harmonization
4. Data wrangling

6



On your mark,
get set, ...



#DoF



The degrees of freedom (#DoF) for a given problem are the number of independent problem variables which must be specified to uniquely determine a solution. Hence the #DoF is important to be considered

1. To design the number of vectors in the learning dataset.
2. To avoid overfitting problem in the model
3. All the above
4. None of the above

A: #DoF



The degrees of freedom (#DoF) for a given problem are the number of independent problem variables which must be specified to uniquely determine a solution. Hence the #DoF is important to be considered

1. To design the number of vectors in the learning dataset.
2. To avoid overfitting problem in the model
3. All the above
4. None of the above

7



On your mark,
get set, ...



Basic metrics in data similarity: cosine



About the cosine metrics it is possible to say that

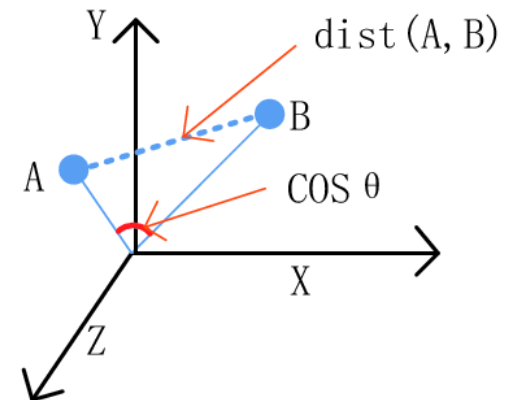
1. Two vectors with the same orientation have a cosine similarity of 1
2. Two vectors oriented at 90° relative to each other have a similarity of 0
3. All of the above
4. None of the above

Basic metrics in data similarity: cosine



About the cosine metrics it is possible to say that

1. Two vectors with the same orientation have a cosine similarity of 1
2. Two vectors oriented at 90° relative to each other have a similarity of 0
3. All of the above
4. None of the above



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On your mark,
get set, ...



Similarity in images datasets



What similarity feature/features discussed in class offers/offer the property to allow a **fast comparison based on a short 1D vector of elements or bits**

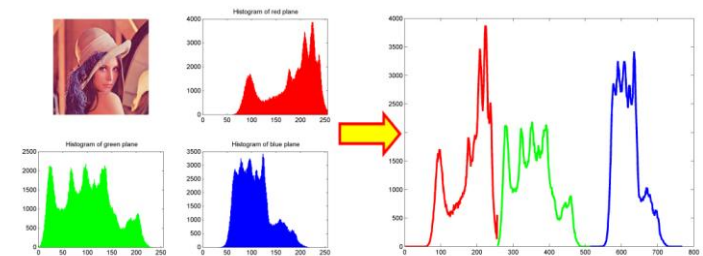
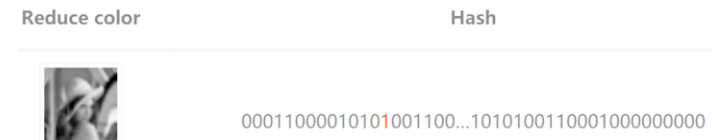
1. phash
2. ahash
3. All the above
4. Cross-correlation

Similarity in images datasets



What similarity feature/features discussed in class offers/offer the property to allow a **fast comparison based on a short vectors of elements or bits**

1. phash
(p = Perceptive ... Discrete Cosine Transform)
2. ahash
(a = Average... 0 if gray > below the mean gray)
3. All the above
4. Cross-correlation
(this is wrong)



Same with histograms comparison

9



On your mark,
get set, ...



Similarity in dataset



In agreement to the class discussion, which description better describes the design activity?

1. Similarity in the dataset requires more space and processing time
2. Similarity in the dataset can improve generalization
3. Both of the above
4. None of the above

Similarity in dataset



In agreement to the class discussion, which description better describes the design activity?

1. Similarity in the dataset requires more space and processing time
2. Similarity in the dataset can improve generalization
3. Both of the above
4. None of the above



We have a «dilemma» between case #1 and #2

10



On your mark,
get set, ...



Dataset preparation: cardinality of checks ops.



In agreement to the class discussion, in a dataset of 1100 labelled images, the search for duplications is typically achieved...

1. by manual exploration of the dataset for better results since the number of images is not critical.
2. by automatic iterations.

Dataset preparation: cardinality of checks ops.



In agreement to the class discussion, in a dataset of 1100 labelled images, the search for duplications is typically achieved...

1. by manual exploration of the dataset for better results since the number of images is not critical
2. by automatic iterations

A basic check $\rightarrow$ $\#iter = N*(N-1) > \mathbf{1.2M}$ comparisons

11



On your mark,
get set, ...



Dataset preparation: labelling errors



In agreement to the class discussion, what kind of labelling error is generally the worst case for the accuracy of the generalization of the model?

ERR1 = Duplications with same labels

EER2 = Duplications with different labels

1. ERR1
2. ERR2
3. ERR1 = EE2

A: Dataset preparation: labelling errors



In agreement to the class discussion, what kind of labelling error is generally the worst case for the accuracy of the generalization of the model?

ERR1 = Duplications with same labels

EER2 = Duplications with different labels

1. ERR1
2. **ERR2**
3. ERR1 = EE2



12



On your mark,
get set, ...



Similarity



According to the class discussion, about the relationship between the operation of cross-correlation and convolution it is possible to say that:

1. They are very similar in meaning and mathematical expression
2. Despite the mathematical expression is similar, the meaning and their use is completely different
3. There is no specific relationship since they are different in meaning and mathematical expressions

A: Similarity

Cross-Correlation:

$$G = h \otimes F$$

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k h[u, v] F[i + u, j + v]$$

Just the minus!

Convolution:

$$G = h * F$$

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k h[u, v] F[i - u, j - v]$$

According to the class discussion, about the relationship between the operation of cross-correlation and convolution it is possible to say that:

1. They are very similar in meaning and mathematical expression
2. Despite the mathematical expression is similar, the meaning and their use is completely different
3. There is no specific relationship since they are different in meaning and mathematical expressions

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On your mark,
get set, ...



Similarity: autocorrelation



According to the class discussion, what is the characteristic of the self-correlation ($O = \text{xcor2}(A, A)$) map produced by a generic image?

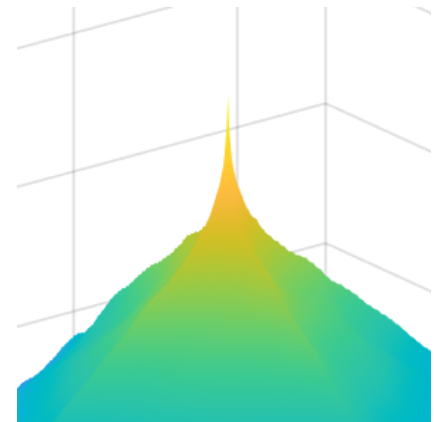
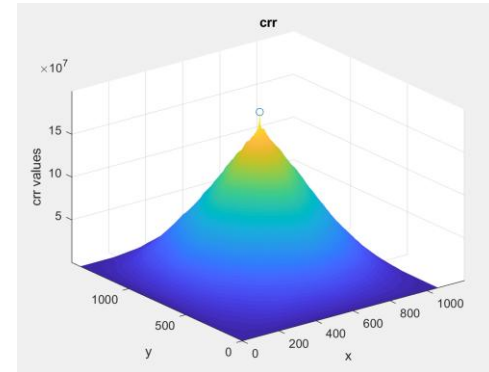
1. A flat and noisy central plateau
2. An evident spike at the center with a very well defined maximum
3. It is not possible to create an autocorrelation map from one single images, two different images are needed

A: Similarity: autocorrelation



According to the class discussion, what is the characteristic of the autocorrelation map produced by a generic image?

1. A flat and noisy central plateau
2. An evident spike at the center with a very well defined maximum
3. It is not possible to create an autocorrelation map from one single images, two different images are needed



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On your mark,
get set, ...



Feature preprocessing/engineering



If your data set contains extreme outliers it better to use as preprocessing

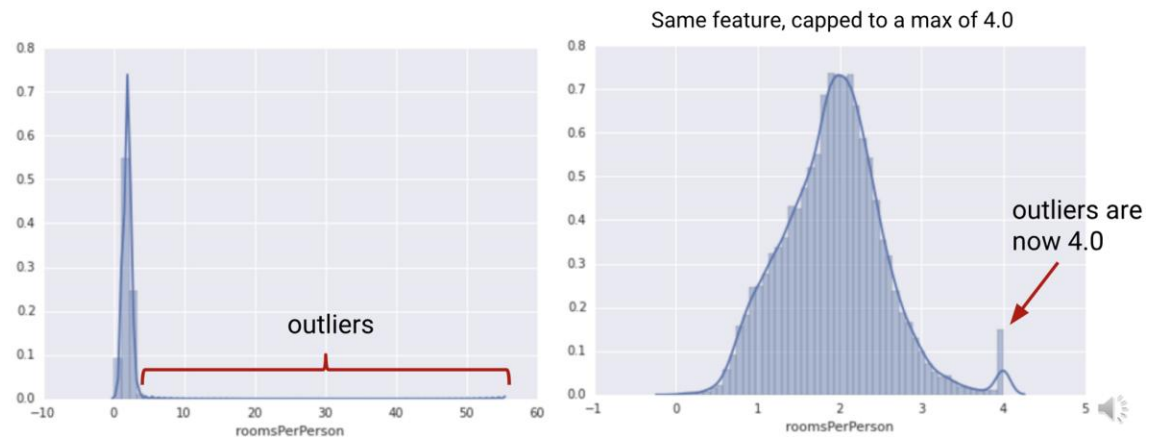
1. Feature clipping
2. Min-max normalization
3. Z' norm

A: Feature preprocessing/engineering



If your data set contains extreme outliers it better to use as preprocessing

1. Feature clipping
2. Min-max normalization
3. Z' norm



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On your mark,
get set, ...



Feature preprocessing/engineering



A logarithmic scaling to one feature values is typically applied in a case of

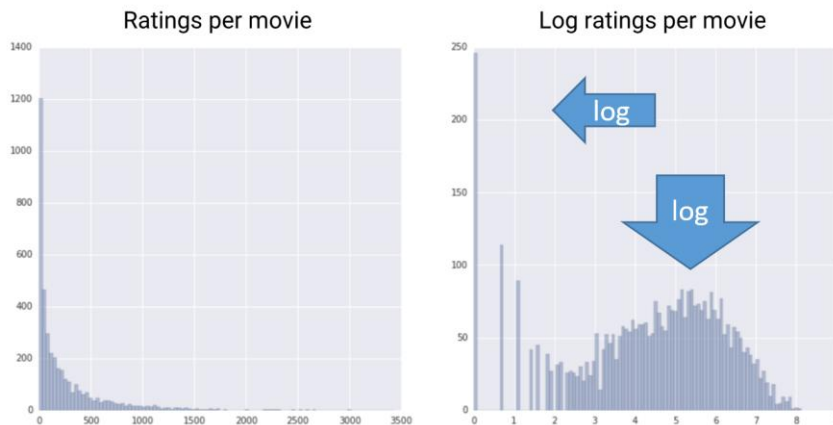
1. Outliers presence
2. Negative values
3. A very large range in the values (>0)

A: Feature preprocessing/engineering



A logarithmic scaling to one feature values is typically applied in a case of

1. Outliers presence
2. Negative values
3. A very large range in the values (>0)



A classical
min-max normalization/z norm
is not effective

16



On your mark,
get set, ...



Plotting of the results



According to the scientific visualization rules presented in class, if you are plotting many figures of merit obtained by your trained neural network on a new dataset, which is the correct ranking of visual attributes to be used?

Left: low accuracy **Right: HIGH ACCURACY**

1. Color intensity > Hue > Length
2. Area > Length > Hue
3. Slope > Angle > Volume
4. Hue > Area > Length

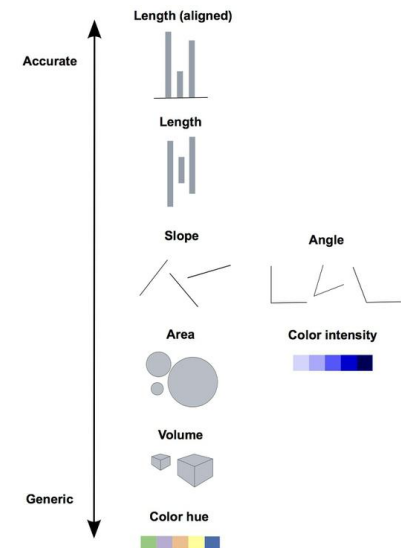
A: Plotting of the results



According to the scientific visualization rules presented in class, if you are plotting many figures of merit obtained by your trained neural network on a new dataset, which is the correct ranking of visual attributes to be used?

Left: low accuracy Right: **HIGH ACCURACY**

1. Color intensity > Hue > Length
2. Area > Length > Hue
3. Slope > Angle > Volume
4. Hue > Area > Length



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On your mark,
get set, ...



Plotting of the results (2)



According to the scientific visualization rules presented in class, is it possible to plot a graphical representation of the confidence level of your figures of merit of your trained model?

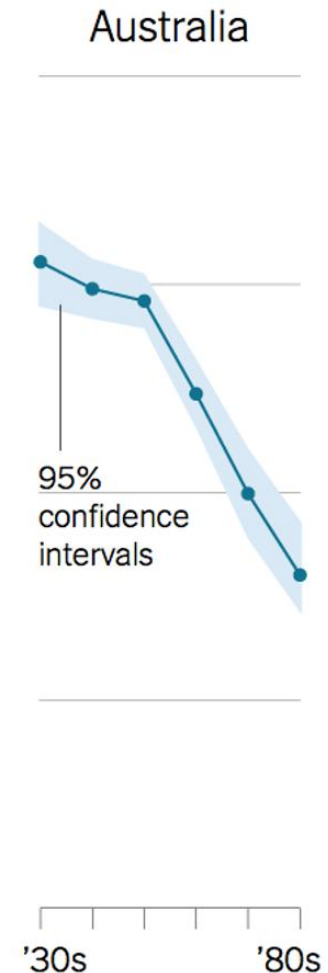
1. No, it is a statistical index with different units and meaning and hence can not be represented in the same plot
2. Yes, the confidence interval data have the same units and meaning and they can be represented in the same plot

A: Plotting of the results (2)



According to the scientific visualization rules presented in class, is it possible to plot a graphical representation of the confidence level of your figures of merit of your trained model?

1. No, the confidence intervals data have different units and meaning and hence can not be represented in the same plot
2. Yes, the confidence interval data have the same units and meaning and they can be represented in the same plot



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On your mark,
get set, ...



Plotting of the results (3)



According to the discussion presented in class about the data visualization, and considering the following steps of the design workflow

1) Get Data, 2) Clean Manipulate Data, 3) Train models, 4) Test Data, 5) Improve the design, which are the main step/steps where data visualization should be involved?

- A. #1
- B. #5
- C. #1 and #5
- D. #2, #3 and #5

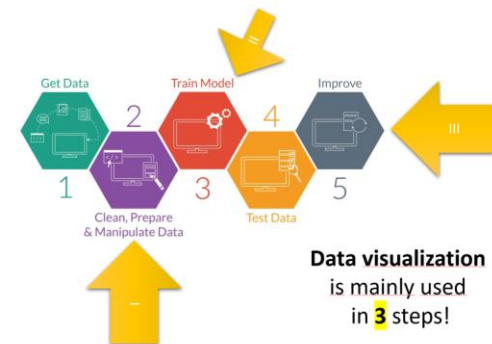
A: Plotting of the results (3)



According to the discussion presented in class about the data visualization, and considering the following steps of the design workflow

1) Get Data, 2) Clean Manipulate Data, 3) Train models, 4) Test Data, 5) Improve the design, which are the main step/steps where data visualization should be involved?

- A. #1
- B. #5
- C. #1 and #5
- D. #2, #3 and #5



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On your mark,
get set, ...



Similarity



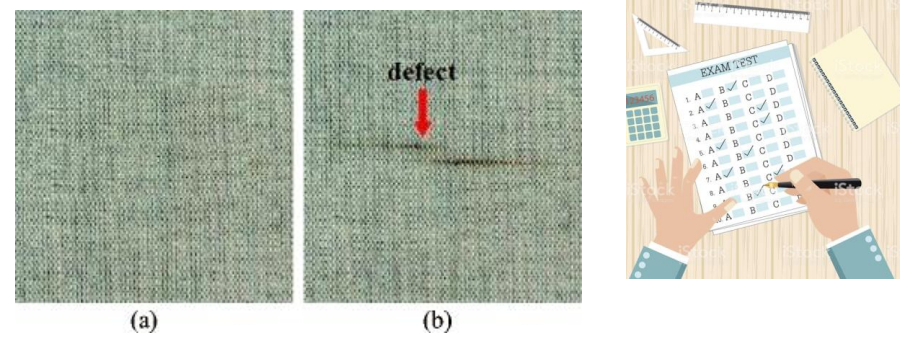
According to the discussion presented in class about the similarity,

Consider an image $A(x,y)$ with internal similarity (repetitions of patterns).

What happens to the output of the self crosscorrelation ($O = \text{xcorr2}(A,A)$)

- A. It is not possible to apply the crosscorrelation to the same image
- B. Output O tends to be a flat plateau with one clear central peak
- C. Output O tends to have many peaks and one evident maximum
- D. Output O tends to have many equivalent peaks with the same maximum value

Similarity

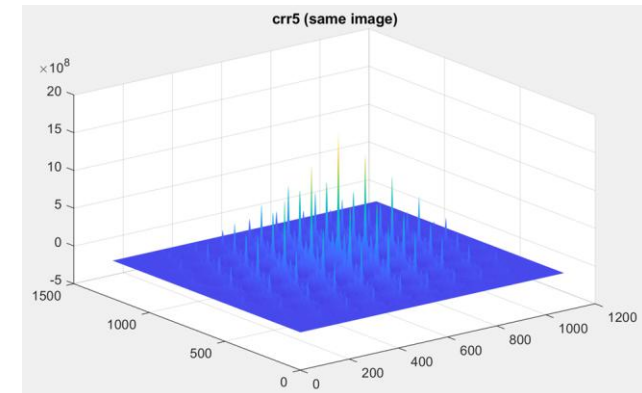
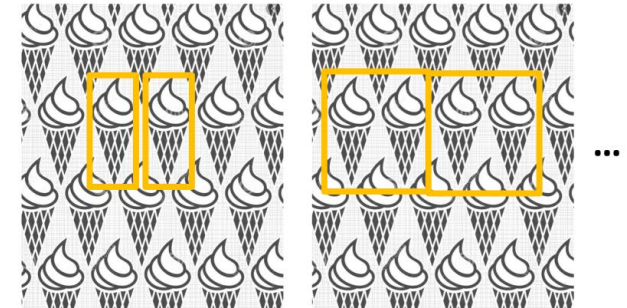


According to the discussion presented in class about the similarity,

Consider an image $A(x,y)$ with internal similarity (repetitions of patterns).

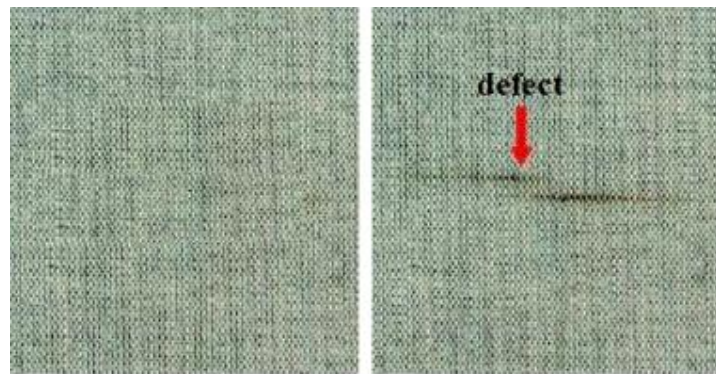
What happens to the output of the self crosscorrelation ($O = \text{xcorr2}(A,A)$)

- A. It is not possible to apply the crosscorrelation to the same image
- B. Output O tends to be a flat plateau with one clear central peak
- C. Output O tends to have many peaks and one evident maximum
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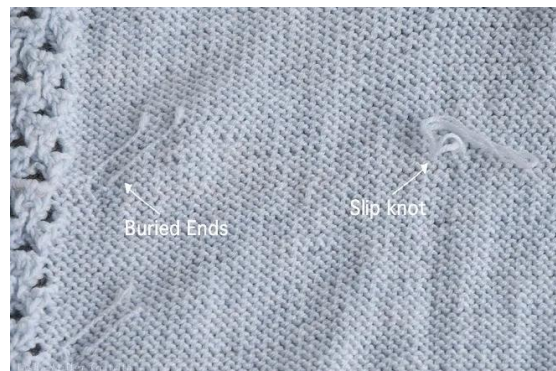
Similarity analysis to find defects in applications

- Defects in applications can be found as differences/lacks in the similarity pattern



(a)

(b)



20



On your mark,
get set, ...



Regulatory framework for AI



According to the discussion presented in class, the EU regulatory framework for AI is:

- A. Mainly focused on public services
- B. Mainly focused on health-related applications
- C. Mainly focused on data privacy
- D. None of the other options

Regulatory framework for AI



According to the discussion presented in class, the EU regulatory framework for AI is:

- A. Mainly focused on public services
- B. Mainly focused on health-related applications
- C. Mainly focused on data privacy

D. None of the other options (since it is quite larger!)

High risk:

- critical infrastructures (e.g. transport), that could put the life and health of citizens at risk;
- educational or vocational training, that may determine the access to education and professional course of someone's life (e.g. **scoring of exams**);
- safety components of products (e.g. AI application in robot-assisted surgery);
- employment, management of workers and access to self-employment (e.g. CV-sorting software for recruitment procedures);
- essential private and public services (e.g. **credit scoring, denying citizens opportunity to obtain a loan**);
- law enforcement that may interfere with people's fundamental rights (e.g. evaluation of the reliability of evidence);
- migration, asylum and border control management (e.g. verification of authenticity of travel documents);

Limited risk

- Limited risk refers to AI systems with specific **transparency obligations**
- When **using AI systems such as chatbots, users should be aware that they are interacting with a machine** so they can take an informed decision to continue or step back.

Minimal or no risk

- This includes **applications such as AI-enabled video games or spam filters**
- The vast majority of AI systems currently used in the EU fall into this category.



**THE
END**



How is your accuracy?

Do not hesitate to contact the teacher in case of doubts



What about topics not present in the simulation?

- They are equally important
 - The absence in the simulation does not mean they are not relevant
- You can find them in the exam
- Use the simulation not to study but to understand you are profitably attending the course and, in case, change the study method
- Try to create your own simulation by browsing all slides of the course and ask yourself: **what can be asked in this slide?**



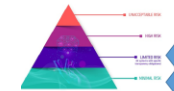
Main points (1)

- Exam simulations are not intended as study material, but instead to TEST your
 - preparation
 - studying style about the course
- Review the course's slides asking in a proactive fashion and by asking yourself...



“What can
be asked
here?”

AI regulation in EU
from high risk to low risk



Limited risk

- Limited risk refers to AI systems with specific transparency obligations
- When using AI systems such as chatbots, users should be aware that they are interacting with a machine so they can take an informed decision to continue or step back.

Minimal or no risk

- This includes applications such as AI-enabled video games or spam filters
- The vast majority of AI systems currently used in the EU fall into this category.

Main points (2)



- Use statistics to your advantage by reasoning not to find the right answer, but **to exclude the wrong ones**

The following activity:

- a) Data Selection;
 - b) Data Filtering;
 - c) Data Enhancing...
- ~~1. Are part of the job of the artificial intelligent specialist in normal activities~~
 - ~~2. Contribute to keep lower the complexity of the learning task~~
 3. All the above (1. and 2.)
 - ~~4. Are part of the classical machine learning approaches and they are (correctly) no longer used in deep learning applications~~